

Tennessee Comprehensive Assessment Program

TCAP

Science
Grade 3
Practice Items



Which statement **best** explains the water that is found on Earth.

- A. It is only a liquid and stays where it is found.
- B. It is only a liquid and moves around the Earth.
- C. It can be a solid, liquid, or gas and stays where it is found.
- D. It can be a solid, liquid, or gas and moves around the Earth.

Students studied types of clouds. The students recorded the types of clouds they saw in the table shown. There were two days of rainfall while students gathered data.

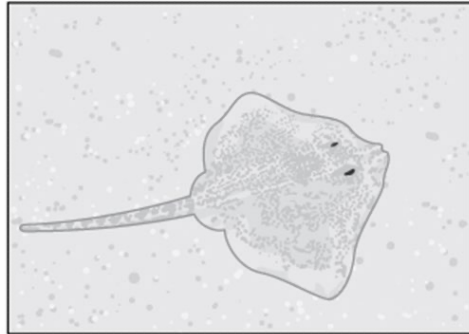
Cloud Types					
Day of the Week	Sunday	Monday	Tuesday	Wednesday	Thursday
Types of Clouds Seen	Cumulus	Cumulonimbus	Cumulonimbus	Cirrus	Cirrus

On which **two** days did rainfall occur?

- M. Sunday
- P. Monday
- R. Tuesday
- S. Wednesday
- T. Thursday

Stingrays have special skin colors and patterns. Students are asked to make a claim about the reason why these special skin colors and patterns help the stingray survive.

Stingray

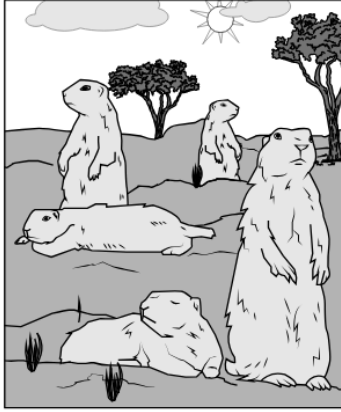


A student says that these special skin colors and patterns help the stingrays avoid predators. Which evidence **best** supports the student's claim?

- A. the size of the stingray
- B. the appetite of the stingray
- C. the ability to stay on the ocean floor
- D. the ability to blend in with the ocean floor

The picture shows some prairie dogs.

Group of Prairie Dogs

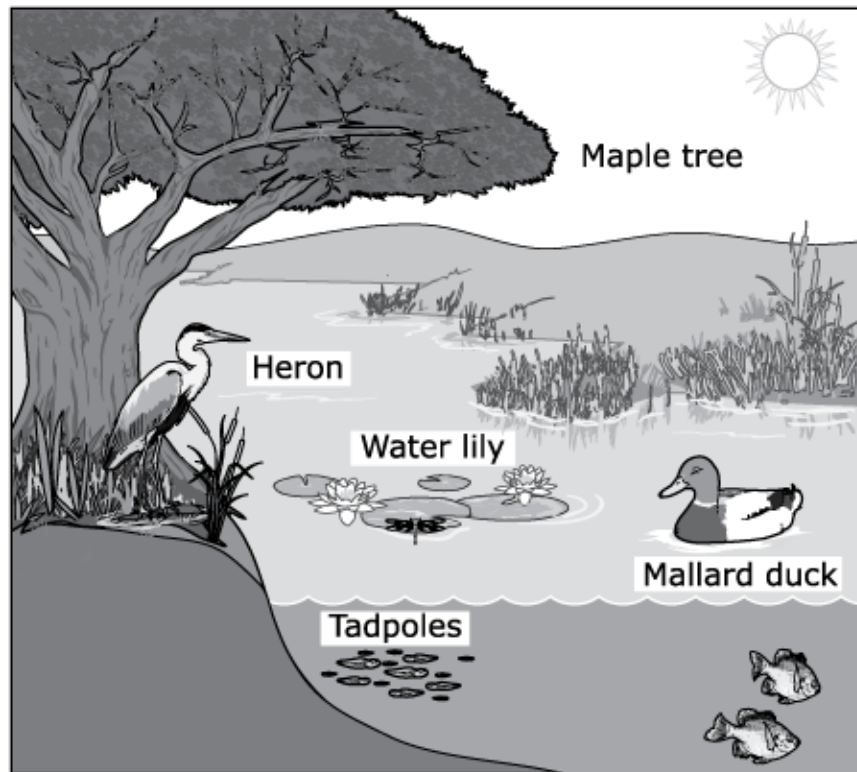


A student claims that living in groups helps prairie dogs.

Which **two** observations of these prairie dogs would support the student's claim?

- M. They are active at night.
- P. They raise their young together.
- R. They are easily seen by predators.
- S. They take turns watching for predators.
- T. They are less selective about what they eat.

A pond is shown. During the hot summer months, the water in the pond gets lower. By the end of the summer, the pond has completely dried up.



Which **three** results are **most likely** due to the loss of the pond?

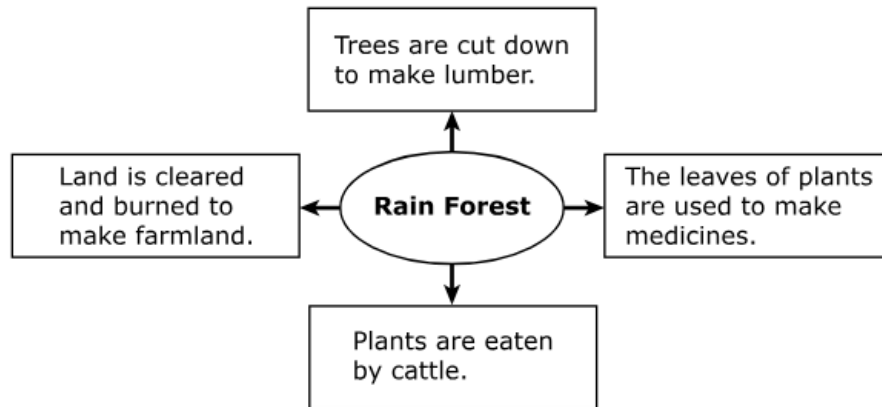
- A. The roots of the maple tree will grow deeper in the soil to find water.
- B. The heron will move to another pond with water.
- C. The water lily will wilt and die in the dry pond.
- D. The fish will move to another pond with water.
- E. The mallard duck will build a nest in the dry pond.

Some places on Earth have a cold climate. These places also have very strong winds. Only a few animals live there. Plants that grow there are usually very short. No tall trees grow in these places.

Which of these is the **most likely** reason why short plants are able to grow in these places?

- M. The roots of short plants have shallow growth in the cold ground.
- P. The leaves of short plants are eaten by animals.
- R. The flowers of short plants become covered by snow in the winter.
- S. The stems of short plants break in the strong winds.

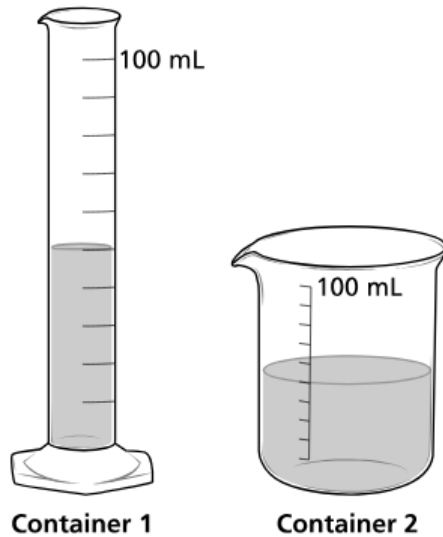
Students look at a diagram of rain forest resources. The diagram shows how rain forest resources are used.



Based on the diagram, if the rain forest is cleared, which natural resource will be **most likely** lost?

- A. types of cattle
- B. lumber used for building
- C. land used for farming
- D. plants used for medicine

A student puts 50 milliliters of water into two different containers, as shown.



Which of these **best** compares the properties of the water in each container?

- M. The volume and shape of the water both stay the same.
- P. The volume of the water changes, but the shape stays the same.
- R. The shape of the water changes, but the volume stays the same.
- S. The volume and shape of the water both change.

A teacher showed four objects to a class. The teacher heated each object until it changed. The data table lists the objects and describes how each object changed.

Student Results

Before Heating	After Heating
Solid butter	Liquid butter
Solid chocolate	Melted chocolate
Raw egg	Cooked egg
Candle wax	Melted wax

Which object's change may **not** be reversed by cooling?

- A. butter
- B. chocolate
- C. egg
- D. wax

A student uses a balance to compare the masses of different fruits. The results are shown.



Which fruit has the **greatest** mass?

M.



P.



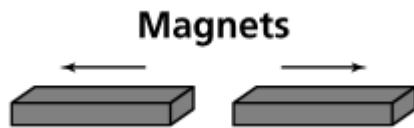
R.



S.



A student places two magnets end to end. The magnets push away from each other.



These magnets push away from each other because

- A. each magnet has a different mass.
- B. each magnet is made of a different material.
- C. the south pole of one magnet is lined up with the south pole of the other magnet.
- D. the south pole of one magnet is lined up with the north pole of the other magnet.

A teacher measured how fast a student threw a baseball. The student threw the ball five times at a plank of wood. The fastest throw damaged the wood the most. The slower the throw, the less damage there was to the wood.

Throwing Speeds

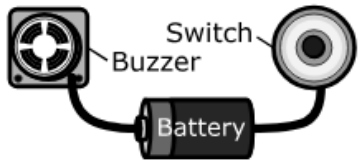
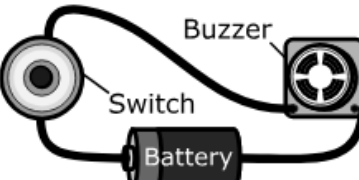
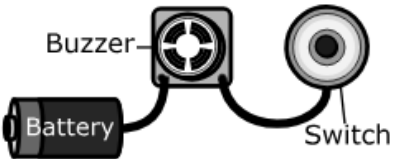
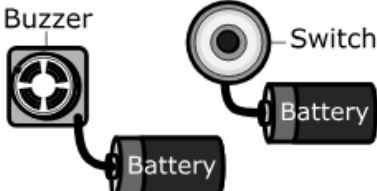
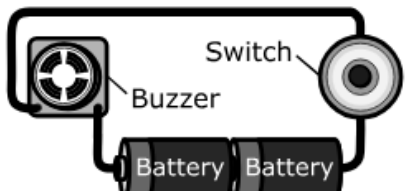
Throw	Ball Speed (miles per hour)
1	35
2	41
3	38
4	43
5	36

Which **two** arguments can students make about the energy of the ball based on the speed and damage to the wood?

- M. Throw 1 had the least energy because it had the slowest speed and damaged the wood the least.
- P. Throw 2 had the most energy because it had the fastest speed and damaged the wood the most.
- R. Throw 3 had the least energy because it had the fastest speed and damaged the wood the most.
- S. Throw 4 had the most energy because it had the fastest speed and damaged the wood the most.
- T. Throw 5 had the least energy because it had the slowest speed and damaged the wood the least.

The table shows some examples of working and nonworking circuits.

Working and Nonworking Circuits

Status	Circuit Design
Nonworking	
Working	
Nonworking	
Nonworking	
Working	

Which statement is **correct** about the working and nonworking circuits in the table?

- A. Working circuits contain closed loops.
- B. Nonworking circuits connect the buzzer and switch.
- C. Nonworking circuits use only one battery.
- D. Working circuits need at least four wires.

Students tested the strength of five magnets. The students put a paper clip on a desk. They slowly pushed each magnet toward the paper clip from far away. They stopped pushing the magnet as soon as the paper clip started to slide toward the magnet. Then they measured the distance between the magnet and the paper clip. They recorded the distance in the data table shown.

Magnet Strength Data

Magnet	Distance (inches)
V	1
W	2
X	2
Y	4
Z	5

Which **two** statements correctly describe these data?

- M. Magnet V has the weakest pull.
- P. Magnet V is stronger than Magnet Z.
- R. Magnet W is as strong as Magnet X.
- S. Magnet Y has the strongest pull.
- T. All of the magnets are the same strength.

Metadata- Science Items

Page Number	Item Type	Key	TN Standards
1	MC	D	3.ESS2.1
2	MS	P, R	3.ESS2.2
3	MC	D	3.LS1.1
4	MS	P, S	3.LS2.1
5	MS	A,B,C	3.LS4.1
6	MC	M	3.LS4.2
7	MC	D	3.LS4.3
8	MC	R	3.PS1.1
9	MC	C	3.PS1.2
10	MC	R	3.PS1.3
11	MC	C	3.PS2.1
12	MS	M, S	3.PS3.1
13	MC	A	3.PS3.2
14	MS	M, R	3.PS3.3
Metadata Definitions			
Item Type	MC = Multiple Choice MS = Multiple Select		
Key	Correct answer		
TN Standards	Primary educational standard assessed.		